Perturbation-Seeking Generative Adversarial Networks: A Defense Framework for Remote Sensing Image Scene Classification

Gong Cheng, *Member, IEEE*, Xuxiang Sun, Ke Li, Lei Guo, and Junwei Han

***Abstract*— The methods for remote sensing image (RSI) scene classification based on deep convolutional neural networks (DCNNs) have achieved prominent success. However, confronted with adversarial examples obtained by adding imperceptible perturbations to clean images, the great vulnerability of DCNNs makes it worth exploring effective defense methods. To date, numerous countermeasures for adversarial examples have been proposed, but how to improve the defensive ability for unknown attacks still to be answered. To address this issue, in this article, we propose an effective defense framework specified for RSI scene classification, named perturbation-seeking generative adversarial networks (PSGANs). In brief, a new training framework is designed to train the classifier by introducing the examples generated during the image reconstruction process, in addition to clean examples and adversarial ones. These generated examples can be random kinds of unknown attacks during training and thus are utilized to eliminate the blind spots of a classifier. To assist the proposed training framework, a reconstruction method is developed. First, instead of modeling the distribution of clean examples, we model the distributions of the perturbations added in adversarial examples. Second, to make a tradeoff between the diversity of the reconstructed examples and the optimization of PSGAN, a scale factor named seeking radius is introduced to scale the generated perturbations before they are subtracted by the given adversarial examples. Comprehensive and extensive experimental results on three widely used benchmarks for RSI scene classification demonstrate the great effectiveness of PSGAN when faced with both known and unknown attacks. Our source code is available at https://github.com/xuxiangsun/PSGAN.**

***Index Terms*— Adversarial defense, generative adversarial networks (GANs), image reconstruction, remote sensing image (RSI) scene classification.**

I. Introduction

DUE to the great development of remote sensing imaging technology, various images with higher resolution (e.g., spatial resolution and spectral resolution) and quality can be accessed, which brings strong requirements to researchers for the understanding of remote sensing images (RSIs). Among the numerous requirements, RSI scene classification is a fundamental task. In short, RSI scene classification aims to classify an RSI into a specific scene category by the subregions extracted from it [1]. For this task, the big within-class diversity and the high between-class similarity make it more challenging to be solved [2].

Over the years, continuous efforts have been made towards how to extract more discriminative features. The early methods mainly focus on handcrafted features, such as texture and color features [3]–[5]. Different from those, mid-level approaches are developed by building more comprehensive representations with the help of high-order statistical methods [6]–[8]. Driven by the rapid development of deep learning algorithms, recent approaches based on deep learning algorithms, especially deep convolution neural networks (DCNNs), have achieved prominent success [9]–[16].

Despite the great success achieved by those DCNN-based methods, they also show great vulnerability when faced with adversarial examples. In brief, the adversarial examples are obtained by adding imperceptible perturbations to original clean images. From the perspective of human, the images after adding perturbations cannot be distinguished from original clean images, but they can be misclassified by excellent DCNNs. Recent research works indicate that adversarial examples can disturb DCNNs in many fields, such as image classification [17]–[21], object detection [22]–[24], and face recognition [25], [26].

Also, for RSI scene classification, adversarial examples still bring great challenges to researchers [1], [20]. Since the vulnerability of DCNNs, there is an urgent need to give further research for defending these adversarial examples to maintain the performance of DCNNs. To date, numerous countermeasures for adversarial examples have been developed [17], [18]. Among those countermeasures, there is still a key flaw to be addressed, that is, how to improve the defensive ability against unknown attacks.

To deal with this issue, in this article, we propose an end-to-end defense method specific for RSI scene classification, named perturbation-seeking generative adversarial networks (PSGANs). Briefly speaking, PSGAN is achieved by introducing the reconstructed examples into adversarial training [27]–[30], which generally uses clean examples and adversarial examples to train a classifier. During the training process of

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image reconstruction, the reconstructed examples can model diverse kinds of adversarial examples, which are further leveraged by adversarial training to build a robust mapping.

To better incorporate the proposed training framework of PSGAN into RSI data, we use generative adversarial networks (GANs) [31] to realize image reconstruction. Recent works for adversarial defense based on image reconstruction [32]–[35] usually model the distributions of clean examples directly. However, the clustered backgrounds in RSI and the instability of GANs make them not suitable since the complicated backgrounds and the instability of DCNNs may make the generator to generate more negative examples for training the classifier. Thus, inspired by the idea of residual networks [36], we model the distributions of the perturbations in adversarial examples. Nevertheless, although more chances will be provided to find more unknown kinds of adversarial examples if the reconstructed examples are acquired by subtracting the generated perturbations from the given adversarial examples, it also brings more challenges for the optimization of PSGAN since a larger starting seeking space will bring more instability for the optimization of methods based on GANs [37]. To make a tradeoff, we propose to multiply the generated perturbation with a scale factor, named seeking radius, before they are subtracted from the given adversarial examples.

Extensive evaluations on three widely used benchmarks (i.e., the UC Merced, Aerial Image dataset (AID), and NWPU-RESISC45 datasets) demonstrate the effectiveness of PSGAN. From the operational viewpoint, our PSGAN can reach higher defensive performance against unknown attacks yet without data preprocessing or modification to the architecture of a classifier, which is more salutary in practice, since unknown attacks are the problem we often face in reality and making modifications to the architecture of a classifier seems more hard for the lack of principle guidance.

The contributions of this article can be summarized as follows.

1) We develop a new training framework by leveraging the reconstructed examples, clean examples, and adversarial examples to train classifiers.
2) We propose a reconstruction method to assist the developed training framework. In detail, we use GANs to model the distributions of the perturbations added to clean examples.
3) To make a tradeoff between the diversity of the reconstructed examples and the optimization of PSGAN, we introduce a scale factor named seeking radius to limit the starting seeking space.

The rest of this article is arranged as follows. In Section II, the related works are briefly introduced. Section III describes our method in detail. Section IV presents the comprehensive results on three benchmarks. Finally, a brief conclusion of this article is summarized in Section V.

II. Related Work

A. Generative Adversarial Networks

GANs [31], a generative model, consist of a generator and a discriminator. The training scheme of GAN can be seen as a two-player mini-max game. The generator tries to generate samples that can fool the discriminator, while the discriminator tries to distinguish real samples from the generated samples.

Inspired by the strong generative ability, GAN has been applied to various vision tasks, especially for image synthesis, such as image-to-image translation [38]–[41], image super-resolution [42], and face editing [43], [44].

In this article, the architecture of our proposed PSGAN is mainly based on Pix2Pix [38], with several modifications, which will be discussed in Section III. Moreover, to overcome the instability of GAN, a two time-scale optimization method, TTUR [45] training strategy, is further applied to PSGAN.

B. Adversarial Attack

In this article, only white-box attacks are considered because the attack transferability of black-box attacks mostly appears on small datasets, according to [34]. A brief review of several representative white-box attacks used in the experiments of this article is presented in the following.

1) Fast Gradient Sign Method: Fast gradient sign method (FGSM) [46] is a kind of gradient-based attack. Also, it is a one-time attack because it only computes gradients once for every image. However, their assumption about the linear nature of the DCNNs makes that FGSM cannot perform well in reality because the existence of activation function in DCNNs makes them not strictly linear to some degree.

Given an image $\boldsymbol{I}$ and its true label $\boldsymbol{y}$, the perturbation δ generated by FGSM is

$$\delta = \epsilon \cdot \text{sign}(\nabla_{\boldsymbol{I}} J(\boldsymbol{I}, \boldsymbol{y})) \tag{1}$$

where $sign(\cdot)$ represents the sign function and $\nabla_{\boldsymbol{I}} J(\boldsymbol{x}, \boldsymbol{y})$ denotes the gradient of an objective function $J(\cdot, \cdot)$ when fed with $\boldsymbol{x}$ and $\boldsymbol{y}$. Besides, with different values of ϵ, which is a coefficient to restrict the norm of the generated perturbation, we can get adversarial examples with different attack strengths.

2) Project Gradient Descent: As for the gradient-based attack, project gradient descent (PGD) [28] can be seen as an iterative version of FGSM. Among various first-order attacks, it is a more powerful attack.

For an image $\boldsymbol{I}$ and its corresponding label $\boldsymbol{y}$, based on the adversarial example $\boldsymbol{I}_t$ at iteration t, the adversarial example $\boldsymbol{I}_{t+1}$ at iteration $t+1$ is calculated by PGD as

$$\boldsymbol{I}_{t+1} = \prod_{I+S} (\boldsymbol{I}_t + \alpha \cdot \text{sign}(\nabla_{\boldsymbol{I}} J(\theta, \boldsymbol{I}, \boldsymbol{y}))). \tag{2}$$

We can get adversarial examples with different attack strengths by setting the parameters α and the attack iterations via PGD.

3) Carlini and Wagner (C&W) Attack: Different from gradient-based attacks, Carlini and Wagner (C&W) [47] is based on optimization. The principle of it is shown in the following equations:

$$\begin{cases} \boldsymbol{d}_n = \dfrac{1}{2}(\tanh(\boldsymbol{w}_n) + 1) - \boldsymbol{x}_n \\ \min\limits_{\boldsymbol{w}_n} ||\boldsymbol{d}_n|| + c \cdot h\left(\dfrac{1}{2}\tanh(\boldsymbol{w}_n) + 1\right) \\ h(\boldsymbol{x}) = \max(\max Z(\boldsymbol{x})_i : i \neq t - Z(\boldsymbol{x})_t, -\kappa) \end{cases} \tag{3}$$

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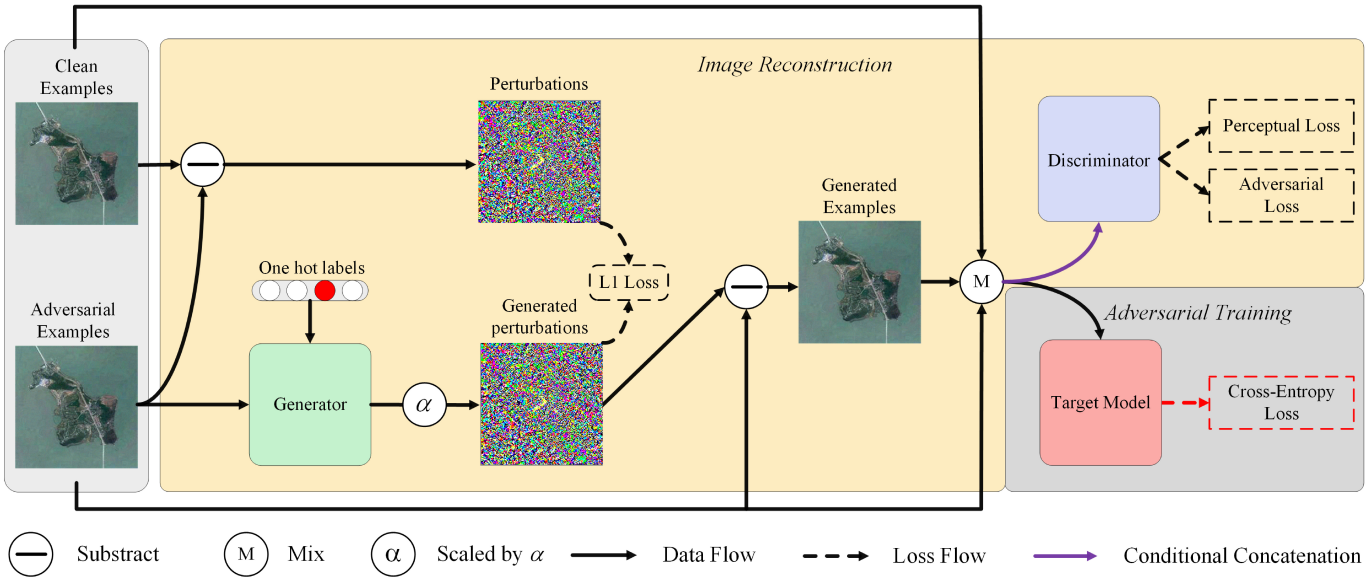


Fig. 1. Training architecture of our PSGAN. The conditional concatenation in this figure means concatenating clean examples with themselves, the input adversarial examples, and the reconstructed examples. The losses represented by the black dotted block are used to optimize the generator, and the loss represented by the red dotted block is used to optimize the target model. For the discriminator, we only optimize it with adversarial loss shown in this figure.

where $Z(\boldsymbol{x})$ denotes the output before the softmax layer of the target model (i.e., the attacked classifier) when fed with $\boldsymbol{x}$. The hyperparameters c and κ, which are customized for different datasets, are related to the final performance.

C. Adversarial Defense

Adversarial training is the first defense method proposed to resist adversarial examples, and it was widely used in many previous works [27], [29], [30], [46]. Its core idea is to use adversarial examples and clean ones to train classifiers. However, the existence of blind spots (low-density regions) for unknown attacks in adversarial training makes it not effective when faced with unknown attacks [48].

Denoising the added perturbations in adversarial examples is another way to resist attacks. Based on GANs, APE-GAN [34] and Defense-GAN [35] are proposed. Both of them use the Euclidean distance in pixel level as a criterion to reconstruct clean examples. The difference between our PSGAN and them is that we model the distributions of the perturbations in adversarial examples instead of modeling the distributions of clean examples directly like them.

The most related work to our method is Rob-GAN [27], which resists adversarial attacks by training the discriminator with the help of diversely generated examples. However, there are still three main differences between Rob-GAN and our PSGAN. First, Rob-GAN is based on AC-GAN [49], which conducts the classification by the trained discriminator. Nevertheless, for PSGAN that is based on Triple-GAN [50], the trained classifiers are used to conduct the classification. Second, Rob-GAN is a two-stage defense framework because it has to fine-tune the discriminator after training with the generated examples and adversarial examples. However, the developed framework in PSGAN is end-to-end. Finally, the generator of Rob-GAN is applied to model the mapping from randomly sampled noises to the unknown adversarial examples. Different from it, the generated examples in PSGAN are realized by the random reconstruction procedure during training, which is based on image-to-image translation.

III. Proposed Method

A. Overview

Fig. 1 shows the training procedure of our proposed method. The training architecture of our method consists of two parts, i.e., the image reconstruction part and the adversarial training part. At each training epoch, we optimize the generator and discriminator first. Then, the adversarial training strategy is further applied to train a target model.

For image reconstruction, conditional input is applied to both the generator and the discriminator. The difference between them is that, for the generator, besides the adversarial examples, it uses one-hot labels as conditional input, whereas for the discriminator, it uses clean examples as conditional input, by which the discriminator can detect whether the reconstructed examples are adversarial examples. After training GAN, the reconstructed examples, the clean examples, and the adversarial examples are leveraged to train the target model. More details will be discussed in the following.

B. Image Reconstruction

1) Perturbation-Seeking Generator: Different from traditional methods for image reconstruction, which model the distributions of clean examples directly, the generator in PSGAN is used to model the distributions of the perturbations added in adversarial examples. Besides, to make a tradeoff between the optimization of PSGAN and the diversity of the reconstructed examples, we propose to multiply the generated perturbation output from the perturbation-seeking generator with a scale

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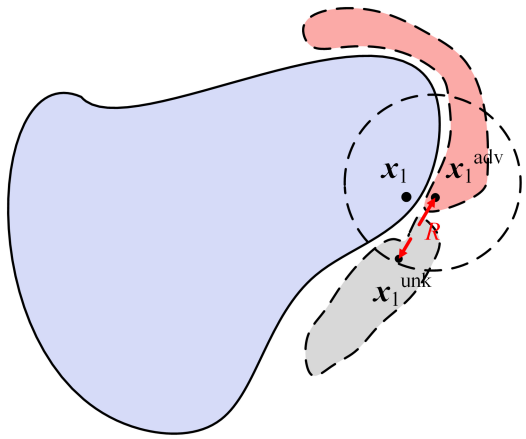


Fig. 2. Illustration of the principle of the perturbation-seeking generator. The polygon surrounded by solid lines represents the distribution of clean examples and the polygon surrounded by dotted lines represents the distribution of adversarial examples. Different colors represent different kinds of examples. $\boldsymbol{x}_1$ belongs to clean examples, $\boldsymbol{x}_1^{\text{adv}}$ belongs to the known adversarial examples, and $\boldsymbol{x}_1^{\text{unk}}$ belongs to the unknown adversarial examples.

factor, named seeking radius. Then, the reconstructed examples can be obtained by subtracting the generated perturbations scaled by the seeking radius from adversarial examples.

A qualitative illustration of our idea is shown in Fig. 2. Let an adversarial example be the center and R be the radius, we can build a starting seeking space, which is a subspace of the whole image space. Then, we can customize the starting seeking range by setting different seeking radii for different target models. Surely, the real starting seeking space may not be a circle, and here, we just use a circle to describe our idea.

To realize what we described above, the proposed perturbation-seeking generator is utilized to model the distributions of the perturbations added in adversarial examples. Then, we use the seeking radius to multiply the generated perturbation output from the generator. This process can be formulated as follows:

$$\boldsymbol{I}_{\text{g}} = \text{Clip}\{\boldsymbol{I}_{\text{adv}} \ominus (\alpha \odot \boldsymbol{P}_{\text{g}})\} \tag{4}$$

where α is the seeking radius. What should be noticed is that the seeking radius represented by R in Fig. 2 actually is $\alpha \odot \boldsymbol{P}_{\text{g}}$ in (4). $\boldsymbol{I}_{\text{adv}}$ is the known adversarial example used to train the generator, $\boldsymbol{P}_{\text{g}}$ is the generated perturbation output from the generator directly, and $\boldsymbol{I}_{\text{g}}$ is the newly generated example. Here, $\ominus$ represents elementwise subtraction, $\odot$ represents the multiplication of scalar and matrix, and $Clip\{\boldsymbol{x}\}$ means to clip the range of the pixel value of each element in $\boldsymbol{x}$ to its predefined interval. In this way, the larger α is, the larger the starting seeking range is. When α is infinitesimal, it means seeking from a very small neighborhood of the known adversarial examples. However, the seeking radius cannot tend to be infinite, because when the pixel value exceeds its predefined interval, it will be clipped to its interval.

As for the architecture of the perturbation-seeking generator, it is similar to [51], while there are two modifications made. First, we add a convolutional block attention module (CBAM) [52] block after the last convolution layer and before the first deconvolution layer respectively. Second, we use one-hot labels as the conditional information fed to the generator. Specifically, the labels are processed to one-hot labels first. Then, they are injected by spatial reshaping and concatenation into the generator together with images. Nevertheless, different from traditional label embedding methods, which only embed labels in the first convolution layer, we also embed labels in the first deconvolution layer.

For the optimization of the perturbation-seeking generator, three objective functions are utilized. The first one is the adversarial loss provided by the discriminator, as shown in the following equation:

$$\mathcal{L}_{\text{adv}}^{\text{G}} = -\mathbb{E}_{\boldsymbol{x},\boldsymbol{x}'}\log(D(\boldsymbol{x},\boldsymbol{x}')). \tag{5}$$

Notice that, $D(\boldsymbol{x},\boldsymbol{x}')$ denotes the logits output from the last layer of the discriminator after concatenating $\boldsymbol{x}$ and $\boldsymbol{x}'$ together as the input to the discriminator. Here, $\boldsymbol{x}$ represents clean examples and $\boldsymbol{x}'$ represents the reconstructed examples, which are obtained by subtracting the generated perturbations scaled by α from the adversarial examples.

Besides, as the goal of our perturbation-seeking generator is to assist adversarial training, thus the reconstructed examples should as "clean" as the original clean images or the known adversarial examples, while the criterion about "clean" should consider two aspects: human visual similarity and CNN visual similarity. For the first one, we use the $L1$ loss as the second objective function of the perturbation-seeking generator, as shown in the following equation:

$$\mathcal{L}_1^{\text{G}} = \mathbb{E}_{\boldsymbol{p},\boldsymbol{p}'}||\boldsymbol{p} - \boldsymbol{p}'||_1 \tag{6}$$

where $\boldsymbol{p} \sim p_{\text{adv-p}}(\boldsymbol{x})$ denotes the perturbations in adversarial examples and $\boldsymbol{p}' \sim p_{\text{g-p}}(\boldsymbol{x})$ denotes the generated perturbations after scaled by α. Moreover, since the discriminator is served as an adversarial example detector in PSGAN, understanding the "visual process" of the discriminator will be more effective for the generator in the improvement of its performance than only knowing the "visual result" of the discriminator. Thus, motivated by pix2pixHD [53], we use the features extracted from the last three layers of the discriminator to help the generator learn the "visual process" of the discriminator, which is shown in the following equation:

$$\mathcal{L}_{\text{Dfeat}}^{\text{G}} = \mathbb{E}_{\boldsymbol{x},\boldsymbol{x}'}||D_{\text{feat}}(\boldsymbol{x},\boldsymbol{x}) - D_{\text{feat}}(\boldsymbol{x},\boldsymbol{x}')||_1 \tag{7}$$

where $\boldsymbol{x} \sim p_{\text{data}}(\boldsymbol{x})$ denotes the clean examples and $\boldsymbol{x}' \sim p_{\text{g}}(\boldsymbol{x})$ denotes the generated examples. What needs to be emphasized is that $\boldsymbol{x}'$ is obtained by subtracting the generated perturbations scaled by α from adversarial examples. $D_{\text{feat}}(\boldsymbol{x},\boldsymbol{x})$ denotes the multiscale features obtained from the last three layers before the last convolution layer of the discriminator, after concatenating $\boldsymbol{x}$ and $\boldsymbol{x}$ together as the input.

In summary, the total loss function used to optimize the generator is shown in the following equation:

$$\mathcal{L}_{\text{G}} = \mathcal{L}_{adv}^{\text{G}} + \lambda_1\mathcal{L}_{\text{Dfeat}}^{\text{G}} + \lambda_2\mathcal{L}_1^{\text{G}} \tag{8}$$

where λ_1 and λ_2 are the weight coefficients of $\mathcal{L}_{\text{Dfeat}}^{\text{G}}$ and $\mathcal{L}_1^{\text{G}}$, respectively.

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TABLE I
CLASSIFICATION RESULTS AGAINST DIFFERENT ATTACKS ON THREE BENCHMARKS

Method	UC Merced		AID		NWPU-RESISC45	
	50%	80%	20%	50%	10%	20%
0-0(A)	96.91±0.47	99.05±0.52	91.08±0.70	94.05±0.73	86.78±0.16	90.28±0.03
0-0(B)	97.83±0.30	99.05±0.33	92.53±0.69	95.66±0.43	89.66±0.15	92.33±0.07
0-0(C)	96.46±0.43	97.32±0.23	90.58±1.32	94.61±0.56	85.47±0.30	89.72±0.23
1-1(A)	28.40±1.47	30.42±0.79	17.42±0.97	17.94±1.22	13.58±0.21	15.74±0.46
1-1(B)	35.67±3.78	38.34±2.62	19.42±5.70	28.56±3.58	15.77±0.41	21.06±0.35
1-1(C)	32.61±2.80	38.57±2.73	19.59±2.03	31.24±1.57	16.44±2.44	21.97±1.38
2-2(A)	22.57±2.24	24.17±2.64	12.58±0.78	14.71±0.98	8.23±0.73	8.94±0.80
2-2(B)	28.15±1.75	29.11±1.62	18.8±1.14	19.74±1.77	9.26±0.86	10.43±1.14
2-2(C)	9.98±2.54	21.97±2.23	9.70±0.56	12.13±0.91	4.34±0.32	5.08±0.52
3-3(A)	21.65±5.92	23.51±3.25	10.87±2.06	11.78±0.91	6.63±3.36	6.27±3.56
3-3(B)	23.66±5.96	26.43±3.14	12.24±2.66	16.64±1.32	5.88±3.48	5.67±3.17
3-3(C)	9.48±9.19	11.94±2.95	1.60±0.19	3.77±0.73	0.35±0.21	0.40±0.22

Method "#-*(X)" means using the adversarial examples generated by # to train model X and test the trained X with *, where 0 for clean examples, 1 for "FGSM", 2 for "PGD", 3 for "C&W", A for ResNet152, B for ResNeXt101, and C for SCNet50. The same below.

2) Discriminator: An Adversarial Example Detector: For the discriminator in PSGAN, the image-conditional input that is the same as pix2pix [38] is applied. The difference between PSGAN and pix2pix is that the discriminator in PSGAN is served as an adversarial example detector, instead of the fidelity detector only. In detail, the clean examples are concatenated with themselves, the input adversarial examples, and the reconstructed examples. Then, these three kinds of concatenated examples are fed into the discriminator. For the criterion of the discriminator, only the clean examples that concatenated with themselves are seen as the True class. In this way, if the reconstructed examples are classified to true class, they should consist of two properties: high fidelity and adversarial property.

Besides, to keep the performance of the perturbation-seeking generator proposed in PSGAN, in addition to the logits output from the last layer, the output also includes the multiscale features obtained from the last three layers of our discriminator. Specifically, the features with different scales are processed to the same scale by a max-pooling layer with stride $= 2$ or stride $= 4$. Then, they are concatenated together after multiplied by fixed multipliers.

The objective function used to optimize the discriminator is shown in the following equation:

$$\mathcal{L}_{\text{adv}}^{\text{D}} = -\mathbb{E}_{\boldsymbol{x}}\log(D(\boldsymbol{x},\boldsymbol{x})) - \mathbb{E}_{\boldsymbol{x},\boldsymbol{x}'}\log\big(1 - D(\boldsymbol{x},\boldsymbol{x}')\big) - \mathbb{E}_{\boldsymbol{x},\boldsymbol{x}_{\text{adv}}}\log(1 - D(\boldsymbol{x},\boldsymbol{x}_{\text{adv}})). \quad (9)$$

In our method, the last layer of the discriminator is a convolution layer without sigmoid. Here, $\boldsymbol{x}_{\text{adv}} \sim p_{\text{adv}}(\boldsymbol{x})$ denotes the adversarial examples. Furthermore, TTUR [45] is incorporated in PSGAN to stabilize the performance of the generator.

C. Adversarial Training

For adversarial training, it rebuilds the mapping of the target model by considering adversarial examples and clean examples together. However, it can only make use of the known type of attacks, so its defensive ability for unknown attacks is relatively poor [48]. However, if we can generate other kinds of adversarial examples and make use of these examples together with known adversarial examples and clean examples to train the target model, then we can strengthen the defense ability for unknown attacks of the target model.

Based on the above analysis, benefiting from image reconstruction, the adversarial training strategy is further applied after generating reconstructed images from the perturbation-seeking generator. During adversarial training, the reconstructed examples, which can be seen as unknown types of adversarial examples, together with clean examples and adversarial examples can be used to fill the blind spots.

Specifically, for adversarial training, we use the cross-entropy loss shown in (10) to optimize the target model, which can also be called the target classifier.

$$\mathcal{H}(\theta,\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{y}\log(f(\boldsymbol{x})) + (1-\boldsymbol{y})\log(1-f(\boldsymbol{x})) \quad (10)$$

where θ denotes the parameters of the target model, $\boldsymbol{x}$ denotes the input images, and $\boldsymbol{y}$ is their true labels.

For the loss function of adversarial training [1], it uses (11) to optimize the target model

$$\mathcal{L} = \mathcal{H}(\theta,\boldsymbol{x},\boldsymbol{y}) + \mathcal{H}(\theta,\boldsymbol{x}_{\text{adv}},\boldsymbol{y}) \quad (11)$$

where $\boldsymbol{x}_{\text{adv}}$ are the adversarial examples obtained from any kind of adversarial attacks given clean examples $\boldsymbol{x}$.

For PSGAN, we use (12) to optimize the target model

$$\mathcal{L}_{\text{CE}}^{\text{C}} = \mathcal{H}(\theta,\boldsymbol{x}_{\text{adv}},\boldsymbol{y}) + \mathcal{H}(\theta,\boldsymbol{x},\boldsymbol{y}) + \mathcal{H}(\theta,\boldsymbol{x}',\boldsymbol{y}). \quad (12)$$

Given adversarial examples $\boldsymbol{x}_{\text{adv}}$, $\boldsymbol{x}'$ denotes the reconstructed examples obtained from the perturbation-seeking generator.

IV. EXPERIMENTS

A. Dataset Descriptions

In this article, three benchmarks are utilized, including the UC Merced dataset [54], the AID dataset [55], and the NWPU-RESISC45 dataset [56].

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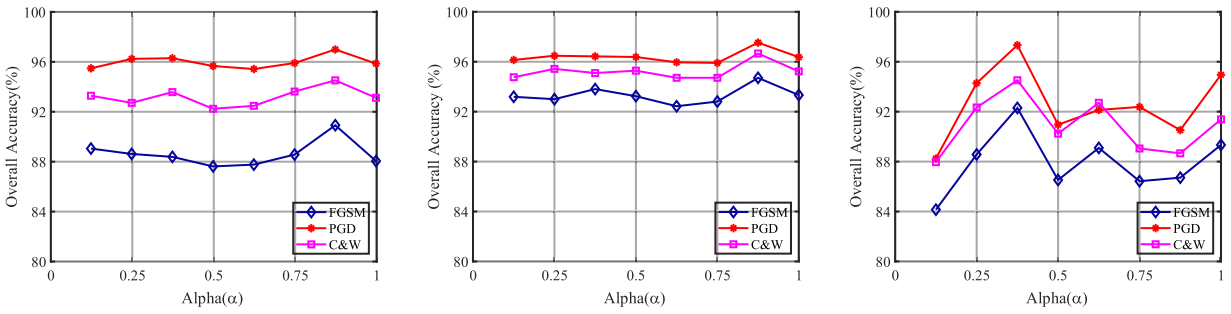


Fig. 3. Overall accuracies of the proposed PSGAN on the UC Merced dataset with 50% as the training ratio under different parameter settings with three target models, which include (a) ResNet152 [36], (b) ResNeXt101 [57], and (c) SCNet50 [58]. The seeking radius α varies from the set of {0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.0}. Here, only the adversarial examples generated by PGD attack are utilized to train these models. What is more, different types of lines shown in the legend represent different types of test examples.

1) UC Merced Dataset: The UC Merced dataset [54] consists of 21 land-use classes. Each scene class contains a total of 100 images measuring 256 × 256 pixels with a resolution of 0.3 m in the red–green–blue (RGB) color space. All these images are selected from the United States Geological Survey (USGS) National Map.

2) AID Dataset: As a large-scale dataset for RSI scene classification, the AID [55] contains 10 000 images altogether, all of them measuring 600 × 600 pixels in the RGB color space are included in 30 scene classes, and each class contains 220–420 images with the resolution varying from 8 to 0.5 m. The images in this dataset are all acquired from Google Earth.

3) NWPU-RESISC45 Dataset: The NWPU-RESISC45 dataset [56] contains 45 scene classes. Each class has 700 images acquired from Google Earth, and the size of all the images is 256 × 256 pixels in the RGB color space. The spatial resolution varies from about 30 to 0.2 m per pixel for most of the 31 500 images. In general, this dataset is the largest in terms of the number of classes and total images. The large within-class diversity and high between-class similarity make the dataset more challenging for RSI scene classification.

For the UC Merced dataset [54], we randomly split it into 80% and 50% for training and the rest 20% and 50% for testing. For the AID dataset [55], the training ratios are 50% and 20%. For the NWPU-RESISC45 dataset [56], we randomly select 20% and 10% of the total images for training and use the rest 80% and 90% of the total images for testing. All the RGB images in these three datasets are resized to 256 × 256 pixels for all experiments in this article. Apart from this, no data augmentation is applied.

B. Experimental Settings

Based on the Pytorch deep learning framework, all of our experiments were conducted on a workstation equipped with ten NVIDIA GeForce GTX2080 Ti GPUs and four Intel Xeon Silver 4114 2.20-GHz CPUs.

For evaluation metric, we use the overall accuracy to quantitatively evaluate the performance of our method. The overall accuracy is defined as the number of correctly classified images divided by the total number of images.

TABLE II
MODEL ABLATION STUDY ON THE UC MERCED DATASET

Method	Adv. Tr. [1]	PSGAN†	PSGAN (**ours**)
2-1(A)	78.59±2.43	85.95±0.88	**88.15±1.79**
2-2(A)	92.59±0.86	93.91±0.45	**95.87±0.31**
2-3(A)	88.21±1.71	91.00±0.88	**93.37±1.34**
2-1(B)	81.41±2.44	90.11±1.00	**93.37±1.09**
2-2(B)	94.00±0.67	94.32±0.61	**96.44±0.60**
2-3(B)	90.72±1.43	91.05±0.22	**95.29±0.61**
2-1(C)	74.67±2.95	85.99±0.94	**89.03±0.69**
2-2(C)	89.26±1.01	90.23±1.11	**94.53±0.77**
2-3(C)	77.89±1.67	87.99±0.87	**90.60±1.70**

PSGAN† stands for modeling the distributions of clean examples instead of the distributions of the perturbations added in adversarial examples. Apart from this, all the other settings in PSGAN† are the same with PSGAN.

Utter experiments can be divided into four main parts. The first part is generating adversarial examples. In this section, first, three target models are fine-tuned on our experimental datasets. Then, we use three kinds of adversarial attacks to obtain adversarial examples. In detail, we use ResNet152 [36], ResNeXt101 [57], and SCNet50 [58] as our target models. For adversarial attacks, we use three typical attacks: FGSM [46], PGD [28], and C&W [47]. During fine-tuning, the batch size of ResNet152 and SCNet50 is set to 32, and for ResNeXt101, the batch size is 16. The learning rate in this part is set to 0.0002 for all these three target models. For the parameters of attack models, we set $\epsilon = 0.02$ for FGSM. For the parameters of PGD, $\alpha = 0.05$ and the attack iteration is equal to 20. For C&W, $c = 0.1$ and $\kappa = 20$. All the adversarial examples are generated offline after the target model is trained.

For the second part, two ablation studies are shown, in order to evaluate the effectiveness of different measures proposed in PSGAN, i.e., the hyperparameter ablation study and the model ablation study. The first one is served as the evaluation of the effectiveness of the introduced scale factor α, and the latter is to evaluate the effectiveness of the modeling method of the perturbation-seeking generator. i.e., modeling the distributions of the perturbations added in adversarial examples.

For the third part, we comprehensively compare the performance of PSGAN with state-of-the-art methods. Since there is only one work is proposed to study adversarial defense for RSI

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TABLE III

CLASSIFICATION RESULTS OF COMPARISON EXPERIMENTS ON THE UC MERCED DATASET. THE HIGHEST ACCURACY IS BOLD-FACED

Method (50%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	88.90±0.72	85.24±0.81	**94.00±0.90**	86.57±2.13	89.57±0.88	**94.76±0.45**	77.64±2.63	88.20±2.43	**90.02±2.04**
1-2	93.64±0.37	90.95±1.07	**96.55±0.25**	93.77±0.78	94.10±0.27	**96.74±0.28**	85.03±3.38	89.29±1.95	**93.16±2.24**
1-3	88.86±0.66	89.29±0.47	**94.21±0.71**	88.76±1.49	93.14±0.00	**94.88±0.78**	68.59±5.96	86.43±1.68	**90.44±2.58**
2-1	78.59±2.43	85.38±0.20	**88.15±1.79**	81.41±2.44	89.72±1.08	**93.37±1.09**	74.67±2.95	88.29±1.75	**89.03±0.69**
2-2	92.59±0.86	90.48±1.75	**95.87±0.31**	94.00±0.67	94.29±0.27	**96.44±0.60**	89.26±1.01	89.62±1.88	**94.53±0.77**
2-3	88.21±1.71	89.34±1.21	**93.37±1.34**	90.72±1.43	93.67±0.47	**95.29±0.61**	77.89±1.67	87.95±1.95	**90.60±1.70**
3-1	76.74±1.76	85.39±0.74	**88.53±1.39**	81.22±1.72	89.29±2.09	**93.20±0.41**	73.03±2.45	86.91±0.33	**89.36±0.83**
3-2	91.60±0.43	89.38±1.41	**95.54±1.12**	93.49±1.03	92.81±0.47	**96.31±0.33**	87.41±0.80	88.34±2.09	**94.26±1.14**
3-3	92.40±0.98	89.67±1.15	**95.01±1.12**	95.01±0.62	92.81±0.07	**96.55±0.42**	88.95±0.74	88.10±1.21	**94.12±1.71**

Method (80%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	94.10±1.00	85.60±1.18	**97.33±1.21**	94.43±1.33	91.31±0.51	**97.52±0.92**	87.19±1.50	91.79±0.50	**94.15±1.98**
1-2	96.19±0.53	93.57±0.00	**98.24±0.99**	98.10±0.48	97.27±1.18	**98.29±0.68**	92.14±1.46	94.17±0.17	**95.33±2.80**
1-3	93.72±0.70	94.05±1.34	**97.33±0.88**	95.24±1.46	96.31±0.85	**97.69±0.54**	83.62±3.13	91.91±0.33	**94.14±2.80**
2-1	85.00±1.12	83.81±0.34	**92.71±1.73**	90.76±2.06	91.19±0.68	**96.86±0.52**	85.76±2.40	90.48±1.68	**94.91±1.31**
2-2	95.89±1.12	93.58±0.67	**97.28±0.80**	97.53±0.64	96.19±1.34	**98.19±0.62**	94.76±1.44	93.81±1.34	**95.99±2.28**
2-3	94.29±1.65	94.64±0.17	**96.61±1.38**	95.38±1.42	95.96±0.67	**98.05±0.87**	87.76±1.67	92.5±1.510	**95.76±1.66**
3-1	84.90±2.36	89.05±2.35	**92.95±0.88**	88.67±1.05	92.86±1.68	**96.53±0.80**	82.14±3.83	90.36±2.19	**95.10±1.31**
3-2	95.43±1.03	94.88±0.51	**97.32±0.69**	96.71±0.78	96.67±1.35	**98.19±0.36**	93.05±0.99	93.10±1.01	**97.15±0.56**
3-3	97.10±0.46	94.29±0.67	**98.15±0.39**	97.90±0.59	97.27±0.84	**98.48±0.64**	94.72±1.45	91.43±0.68	**96.43±1.68**

APE stands for APE-GAN [34]. The data in the brackets under Method represents the training ratios, and the experiments under different training ratios are seperated by double underscore. The same below.

scene classification so far, thus we only compare our PSGAN with adversarial training studied in [1] and a representative defense method for natural images, i.e., APE-GAN [34]. For the experiments in this part, the hyperparameters in (8) are $\lambda_1 = 100$ and $\lambda_2 = 50$, and α in (4) is set to 0.9. The learning rate for the generator of PSGAN is set to 0.0001, and for the discriminator, it is set to 0.0004.

Finally, qualitative results are shown. Specifically, to explore what had changed to a classifier, we visualize the attention maps before the last full-connected layer of a classifier.

The total experimental results are obtained by randomly repeating all the experiments ten times to calculate the average value and standard deviation. In each repetition, we randomly generate different clean training set ten times from the whole dataset with a certain training ratio for different datasets and then carry out all the experimental items.

C. Adversarial Attacks

To evaluate the performance, we have to acquire adversarial examples first. These experimental results conducted on three datasets are summarized in Table I.

From Table I, three conclusions can be established. First, the attack performance of C&W is the greatest than other attacks for different target models, and the performance of FGSM is the worst. Besides, for the same target models and attacks, the more categories of datasets are, the higher performance of adversarial examples will be. Moreover, for attacks on the same dataset, the worse the performance of the target model is, the better the performance of adversarial examples will be.

D. Ablation Studies

All the experiments in this section are conducted on the UC Merced dataset with 50% as its training ratio. Only adversarial examples generated by PGD attack are utilized as the known attack to train the classifiers. Moreover, to comprehensively evaluate the defensive ability for unknown attacks, not only the overall accuracy against adversarial examples generated by PGD attack is reported, but also the accuracy against adversarial examples generated by other attacks are reported. All the other hyperparameters in this section are the same as those reported in Section IV-B.

1) Hyperparameter Ablation Study: In this section, we conduct a simple experiment to quantitatively analyze the impact of different values of the seeking radius on the final classification results. The seeking radius α varies from the set of {0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.0}. The experimental results in this section are reported in Fig. 3.

From Fig. 3, the experimental results imply that the optimal seeking radius for different target models is different. For ResNet152 [36] and ResNeXt101 [57], the relatively optimal seeking radius appears near 0.875, whereas for SCNet50 [58], it appears near 0.375. When α achieves the optimal value in the $(0, 1]$ interval, the overall classification accuracy of these models can exceed the accuracy when $\alpha = 1$ by almost two percentage points. Thus, setting the seeking radius to 1 directly for all target models is not suitable. As we argued in Section III, a larger seeking space will bring more challenges to the optimization of PSGAN although it can provide more chances to find diverse adversarial examples. From this

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TABLE IV

CLASSIFICATION RESULTS OF COMPARISON EXPERIMENTS ON THE AID DATASET. THE HIGHEST ACCURACY IS BOLD-FACED

Method (20%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	78.70±1.50	70.31±1.09	**83.78±0.85**	78.51±2.12	75.95±0.21	**86.54±1.07**	75.61±3.40	74.78±1.67	**81.98±2.70**
1-2	86.33±1.19	79.39±1.48	**88.05±0.34**	87.88±2.03	83.44±1.19	**90.27±0.55**	84.67±2.06	77.61±1.28	**85.78±2.77**
1-3	77.48±0.97	75.95±1.70	**80.14±3.94**	80.39±4.08	80.01±0.74	**86.82±0.81**	77.54±3.48	72.32±1.02	**81.59±4.32**
2-1	66.00±2.12	70.50±0.34	**75.31±0.76**	69.50±3.14	73.07±3.30	**84.85±0.26**	68.37±4.37	74.91±0.86	**77.67±3.17**
2-2	85.05±0.63	79.65±1.41	**87.18±0.64**	86.64±0.95	81.05±0.58	**90.25±0.45**	84.45±0.91	77.69±0.08	**85.37±3.15**
2-3	79.48±0.92	76.43±1.61	**80.59±2.69**	81.20±1.75	77.90±0.48	**86.13±1.33**	79.04±0.89	72.22±0.61	**79.31±3.89**
3-1	56.21±0.98	70.60±2.38	**74.93±0.26**	60.92±1.11	74.42±1.36	**84.18±0.47**	62.35±3.85	69.41±0.66	**81.71±1.00**
3-2	80.25±0.53	74.76±2.44	**86.40±0.54**	82.88±0.29	77.53±2.69	**89.64±0.35**	81.67±2.25	70.56±0.06	**88.37±0.49**
3-3	85.14±0.84	77.40±2.18	**87.32±0.59**	86.93±1.09	80.29±2.96	**89.47±0.34**	84.77±1.66	70.41±1.03	**87.18±0.65**
Method (50%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	89.01±0.47	65.74±4.33	**91.34±0.88**	88.87±1.51	82.75±7.37	**91.86±0.49**	87.69±1.58	79.40±2.21	**90.53±1.26**
1-2	92.35±0.31	83.95±1.09	**93.34±0.49**	92.92±0.67	84.96±6.87	**93.62±0.27**	91.76±1.37	86.72±1.22	**92.41±1.19**
1-3	86.34±1.80	81.21±1.03	**89.22±1.15**	89.09±0.62	84.99±1.20	**90.19±1.67**	88.58±1.11	82.44±0.76	**89.80±0.59**
2-1	79.46±1.34	60.90±1.19	**86.99±0.73**	82.44±1.23	81.53±8.10	**89.97±1.67**	81.60±2.77	78.97±2.81	**89.40±0.97**
2-2	91.14±0.34	82.58±0.54	**91.55±2.18**	92.33±0.34	83.99±7.28	**93.47±0.70**	91.26±0.46	86.36±1.53	**92.86±0.68**
2-3	86.98±1.42	80.20±0.17	**88.44±2.97**	88.32±0.82	85.10±1.30	**90.33±0.71**	87.27±0.80	82.39±1.26	**90.15±0.93**
3-1	70.48±2.01	76.11±8.47	**84.42±1.26**	77.79±1.90	82.54±0.62	**88.54±2.39**	78.10±2.87	80.62±1.87	**87.44±1.97**
3-2	88.28±0.67	85.17±3.80	**91.73±1.16**	90.40±0.89	87.27±0.83	**92.62±0.38**	89.65±0.80	84.05±1.74	**90.52±2.46**
3-3	91.04±0.50	86.68±3.05	**92.12±0.80**	92.44±0.30	86.60±3.17	**92.67±0.55**	**91.40±0.44**	84.71±2.39	88.35±3.00

TABLE V

CLASSIFICATION RESULTS OF COMPARISON EXPERIMENTS ON THE NWPU-RESISC45 DATASET. THE HIGHEST ACCURACY IS BOLD-FACED

Method (10%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	74.47±3.01	50.47±5.21	**79.93±1.71**	74.15±2.77	58.81±5.32	**81.62±1.83**	71.41±2.83	64.45±7.64	**79.99±1.60**
1-2	82.65±2.41	72.97±2.69	**85.39±1.05**	83.85±2.63	78.97±1.80	**86.07±1.25**	81.16±1.81	74.37±0.95	**83.80±2.15**
1-3	71.20±3.01	70.33±1.90	**75.42±2.91**	73.98±3.39	76.24±1.04	**77.81±2.96**	72.34±3.83	69.72±1.10	**79.14±1.12**
2-1	45.55±1.36	49.82±8.85	**67.86±1.96**	49.90±1.11	59.56±8.48	**85.00±1.03**	53.65±3.14	65.23±5.00	**76.22±3.44**
2-2	75.52±0.39	72.69±2.43	**82.93±0.73**	83.65±0.10	78.96±0.71	**86.50±0.62**	79.84±1.00	74.46±1.19	**85.23±1.86**
2-3	81.89±0.21	70.73±1.44	**84.03±0.60**	83.16±0.25	76.42±0.09	**84.72±0.72**	72.53±2.67	69.47±1.00	**78.56±2.80**
3-1	45.55±1.37	60.28±1.30	**67.86±1.96**	49.90±1.11	67.11±2.61	**85.00±1.03**	47.34±3.76	69.37±1.38	**77.17±3.55**
3-2	75.52±0.39	73.94±0.81	**82.93±0.73**	77.84±0.30	**79.02±0.37**	77.30±0.85	75.85±2.35	73.97±0.75	**83.88±2.03**
3-3	81.89±0.20	75.69±2.27	**84.03±0.60**	83.16±0.25	80.11±2.01	**84.72±0.72**	79.32±2.05	73.44±1.90	**82.09±3.17**
Method (20%)	ResNet152 [36]			ResNeXt101 [57]			SCNet [58]		
	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN	Adv. Tr. [1]	APE [34]	PSGAN
1-1	82.99±2.45	36.40±1.87	**85.62±1.23**	82.70±2.19	50.41±3.35	**86.22±1.13**	82.77±2.68	56.91±5.87	**87.24±1.71**
1-2	88.10±1.61	71.42±1.33	**89.06±0.63**	88.83±1.63	79.34±0.50	**89.46±0.55**	87.78±2.36	79.09±1.89	**89.45±1.11**
1-3	76.32±3.41	73.33±2.31	**78.29±2.52**	79.12±3.19	79.42±2.16	**82.33±1.79**	80.48±3.87	76.74±2.80	**83.45±1.95**
2-1	70.38±2.09	37.59±4.47	**79.98±0.41**	74.63±1.56	61.38±7.24	**84.33±0.63**	73.20±3.81	55.00±4.91	**84.96±0.77**
2-2	86.79±0.66	71.22±2.36	**88.87±0.52**	88.03±0.73	75.82±5.95	**89.59±0.45**	86.98±0.74	78.41±1.60	**89.63±0.93**
2-3	77.36±0.52	73.54±2.23	**79.99±0.62**	78.24±1.44	64.91±7.81	**81.33±0.28**	81.42±1.65	76.90±2.35	**84.37±2.02**
3-1	53.94±1.15	46.96±3.52	**79.99±0.26**	60.37±1.29	59.84±6.39	**85.23±2.43**	58.07±0.40	69.43±1.73	**83.32±1.47**
3-2	81.87±0.45	74.84±1.15	**87.39±0.25**	84.13±0.50	81.24±2.40	**87.15±2.00**	82.92±0.97	80.01±0.03	**88.30±0.95**
3-3	86.53±0.42	80.84±0.42	**86.73±0.40**	87.62±0.34	84.90±1.02	**88.47±0.56**	86.05±0.75	81.38±0.74	**88.64±0.77**

perspective, PSGAN can be more flexible for different target models.

2) Model Ablation Study: In this section, a comprehensive model ablation study is conducted since one of our contributions has comprehensively investigated in the hyperparameter ablation study. Thus, we mainly conduct the ablation study to investigate the effectiveness of our modeling method in this section. The experimental results in this section are summarized in Table II. Specifically, for the ablation items in Table II, PSGAN† stands for a modification in PSGAN, i.e., replacing the modeling method of PSGAN. Specifically, the generator in PSGAN† is served to model the distributions of the clean examples. Thus, the reconstructed examples are the output of PSGAN†. Besides, the $L1$ loss used in PSGAN†

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should be the distance between the reconstructed examples and the clean examples. Apart from what we emphasized above, all the other settings are the same as PSGAN.

From Table II, two conclusions can be established. First, compared with adversarial training only, the experimental results of PSGAN and PSGAN† demonstrate that incorporating the reconstructed examples together with clean examples and adversarial examples, and the defensive ability for unknown attacks can be enhanced greatly. As we argued in Section III, the main reason why adversarial training only cannot defense unknown attacks is that there are numerous blind spots in adversarial training. Hence, when faced with unknown attacks, adversarial training will show its vulnerability. However, with the proposed training framework, more kinds of adversarial examples can be explored during the training process of image reconstruction. Once the classifier has seen those adversarial examples, they will learn to build a more robust mapping.

Second, facing the known and unknown attacks, the improvements of PSGAN† are relatively lower than those of PSGAN. As we argued in Section I, the special properties of RSI will bring challenges for the optimization of GAN. Experimental results indicate that modeling the distributions of the perturbations is properly compatible with our proposed training framework.

E. Adversarial Defense

In this section, we compare PSGAN with two representative state-of-the-art methods, i.e., the adversarial training explored in [1] and APE-GAN [34]. To give a better description of the defensive ability for unknown attacks, we not only evaluate the classification results against the known adversarial examples used to train the target model but also evaluate the classification results against unknown adversarial examples.

The experimental results are summarized in Tables III–V. From these tables, we can find two facts. First, facing adversarial examples, the experimental results of PSGAN exceed the results mentioned in [1] by a significant margin on these three benchmarks, especially when faced with unknown attacks. These results effectively illustrate that with the association of the reconstructed examples, PSGAN can reach higher defensive performance for unknown attacks. As we argued in Section III, the poor performance for unknown attacks of adversarial training [1] is because there are numerous blind spots for unknown adversarial examples in adversarial training.

Second, the experimental results of APE-GAN [34] demonstrate that its defensive ability is much lower than adversarial training [1] and our PSGAN. Also, the standard deviations of its results are relatively larger. These facts indicate that the reconstruction method of APE-GAN [34] is not sensitive to the key points of adversarial examples, as argued in [37].

F. Qualitative Results

In this section, a qualitative description about our PSGAN is shown in Fig. 4. In this section, first, adversarial examples generated by FGSM [46] are applied to train the ResNeXt101 [57] by adversarial training [1] and PSGAN. Also, these examples

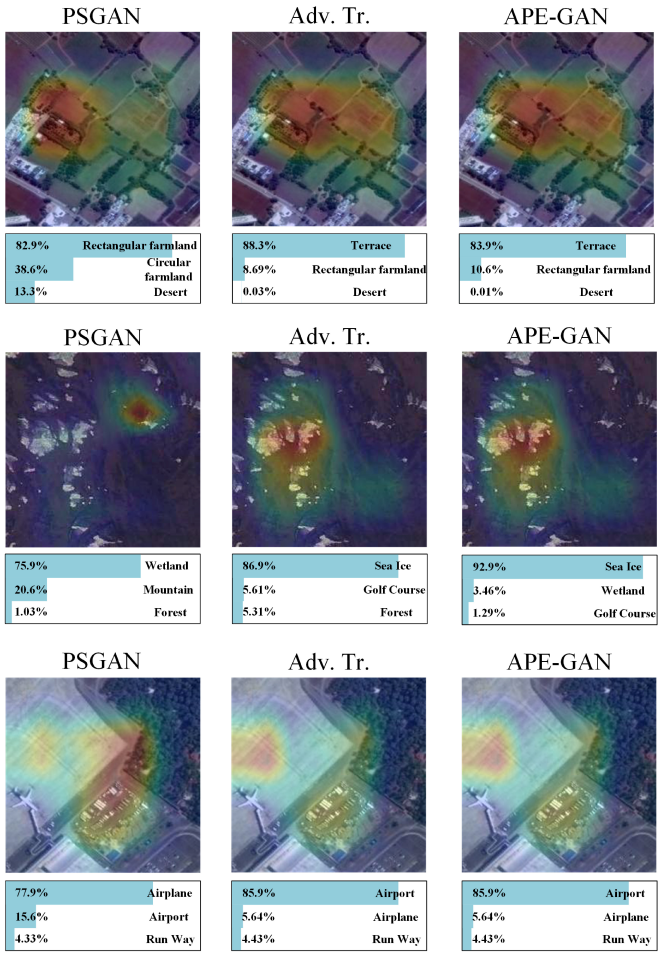


Fig. 4. Attention maps of three adversarial examples extracted from PSGAN, Adv. Tr. [1], and APE-GAN [34]. All these three adversarial examples are generated by FGSM [46] on the NWPU-RESISC45 dataset, and the target model is ResNeXt101 [57]. The blue bars at the bottom of each example show the top three predicted categories and probabilities.

are used to train APE-GAN. Then, we visualize the attention maps before the last full-connected layer of the trained classifier. It is noticeable that since the training framework of APE-GAN [34] does not include the training for the classifiers, so the ResNeXt101 fine-tuned on the NWPU-RESISC45 dataset is chosen to visualize the attention maps of the examples reconstructed by APE-GAN.

From Fig. 4, we can find that not only the adversarial examples can be classified correctly after training by PSGAN, but also the attention maps can focus on specific areas of a particular category. However, for adversarial training [1] and APE-GAN [34], the performances are poorer.

V. Conclusion

In this article, we proposed a simple but effective method to defense adversarial examples in the field of RSI scene classification, named PSGAN. Briefly speaking, a new training framework was proposed to train the classifier by introducing the reconstructed examples, in addition to clean examples and adversarial examples. Besides, to assist the proposed training framework, a new reconstruction strategy was developed

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by modeling the distribution of the perturbations added in adversarial examples. Moreover, a scale factor named seeking radius was applied to scale the generated perturbations before they were subtracted from adversarial examples. Experimental results on three widely used datasets demonstrated the effectiveness of our PSGAN for RSI scene classification when facing both known and unknown attacks.

For future works, there are still a lot of questions to be answered since there are urgent needs to resist the harm of adversarial examples to DCNNs. For example, from the perspective of the defensive ability improvement, whether generating more diverse adversarial examples in our proposed defense framework can result in a higher performance or only a few specific kinds of generated adversarial examples can be utilized to enhance the defensive performance and how to give a proper definition of these kinds of adversarial examples.

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